

```
[1] "Foo"
```

A symbol is a name for another object in R. It is commonly a name of a variable used in an assignment statement. You can identify the type of an R object using the *typeof* function:

```
> typeof(3.1415)
[1] "double"
```

```
> typeof(5)
[1] "double"
```

```
> typeof(0x00FF)
[1] "double"
```

```
> typeof("Foo")
[1] "character"
```

```
> typeof('Foo')
[1] "character"
```

All comments in R begin with the hash (#) symbol. Examples of a single line and multi-line comment are given below:

```
> 3.1415 # Pi
[1] 3.1415
```

```
> # This is a multi-line
> # comment example
```

A variable is assigned a value and you can use the '<-' assignment operator in R:

```
> a <- 1
```

```
> a
[1] 1
```

There are a few special values used in R:

- *Inf* and *-Inf* represent positive and negative infinity respectively.

```
> 10^2048
[1] Inf
```

```
> -10^2048
[1] -Inf
```

- NA refers to 'Not Available', and can be seen when you load text or data into R that has missing values.

- *NULL* is a symbol in R that represents a null object.
- *NaN* means 'Not a Number' and is observed in certain computations like division by zero.

```
> 0 / 0
```

```
[1] NaN
```

Operations

You can perform operations in R such as arithmetic, logical, bitwise, relational, assignment, etc. A few examples of mathematical operations are given below:

```
> 1 + 2
```

```
[1] 3
```

```
> 1 - 2
```

```
[1] -1
```

```
> 1 * 2
```

```
[1] 2
```

```
> 1 / 2
```

```
[1] 0.5
```

```
> 2 ^ 5
```

```
[1] 32
```

A few comparison operator examples are shown in the following R console session:

```
> 1 < 2
```

```
[1] TRUE
```

```
> 1 > 2
```

```
[1] FALSE
```

```
> 1 >= 1
```

```
[1] TRUE
```

```
> 1 <= 1
```

```
[1] TRUE
```

```
> 1 == 1
```

```
[1] TRUE
```

```
> 1 != 2
```

```
[1] TRUE
```

Examples of logical operations are given below:

```
> TRUE & FALSE
```

```
[1] FALSE
```